

Exercise #3

1. Minimum Spanning Trees

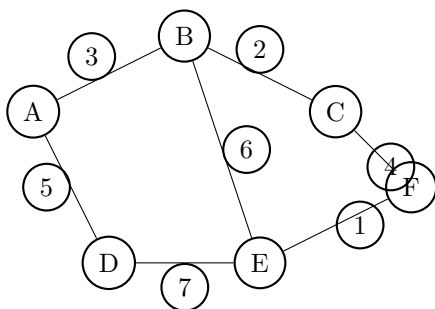
A **minimum spanning tree (MST)** of a connected, weighted, undirected graph is a subset of the edges that:

- connects all vertices together,
- forms no cycles,
- and has the minimum possible total edge weight among all spanning trees.

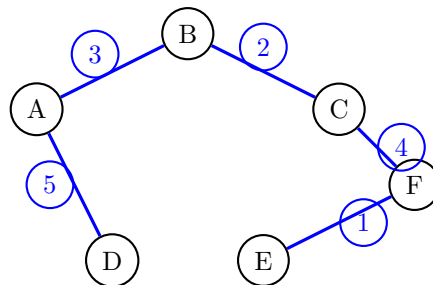
In other words, given a graph $G = (V, E)$ with edge weights $w(e)$, the MST is a tree that spans all vertices V such that the sum of edge weights $\sum_{e \in T} w(e)$ is minimized.

Example Graph

Below is an example of a small weighted graph with six vertices.



(a) Original weighted graph



(b) Minimum spanning tree (total weight = 15)

Figure 1: Example of a weighted graph and its MST.

Explanation

The minimum spanning tree connects all six vertices using the smallest possible total edge weight. In this example, the MST includes the edges with weights **1, 2, 3, 4, and 5**, giving a total weight of $1+2+3+4+5 = 15$.

This MST could be found by using **Kruskal's algorithm**, which:

1. Sorts all edges in increasing order by weight.
2. Adds edges one by one, skipping any that would create a cycle (detected using the Union-Find structure).
3. Stops when $V - 1$ edges have been added.